Enhancing the Efficacy of Teacher Incentives through Loss Aversion: A Field Experiment*

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Abstract

Financial incentives for teachers have not proven effective at increasing student achievement in the United States. In this paper, we replicate that null result using the traditional approach, but also demonstrate that one small tweak to the program – paying teachers their expected bonus *in advance* and taking the money back if students do not improve sufficiently – yields math test score increases between 0.2 and 0.4 standard deviations. This effect is on par with the impact of increasing teacher quality by more than one standard deviation. The results are consistent with loss aversion induced behavioral changes triggered by the upfront payments.

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I. Introduction

Good teachers matter. A one-standard deviation improvement in teacher quality translates into annual student achievement gains of 0.15 to 0.24 standard deviations (hereafter σ) in math and 0.15σ to 0.20σ in reading (Rockoff, 2004; Rivkin et al, 2005; Aaronson et al., 2007; Kane and Staiger, 2008). These effects are comparable to reducing class size by about one-third (Krueger, 1999). Similarly, Chetty et al. (2012) estimate that, a one-standard deviation increase in teacher quality in a single grade increases earnings by about 1% per year; students assigned to these teachers are also more likely to attend college and save for retirement, and less likely to have children when teenagers.

It has proven difficult, however, to identify *ex ante* the most productive teachers. Observable characteristics such as college-entrance test scores, grade point averages, or major choice are not highly correlated with teacher value-added on standardized test scores (Aaronson et al., 2007; Rivkin et al., 2005; Kane and Staiger, 2008; Rockoff et al., 2011). And, programs that aim to make teachers more effective have shown little impact on teacher quality (see e.g., Boyd et al., 2007 for a review). Some argue that these two facts, coupled with the inherent challenges in removing low performing teachers due to collective bargaining agreements and increased job market opportunities for women, contributes to the fact that teacher quality and aptitude has declined significantly in the past 40 years (Corcoran et al., 2004; Hoxby and Leigh, 2004).[1]

To increase teacher productivity, there is growing enthusiasm among policy makers for initiatives that tie teacher incentives to the achievement of their students. At least ten states and many more school districts have implemented teacher incentive programs in an effort to increase student achievement (Fryer, forthcoming).[2] Yet, the

[1]Corcoran et al. (2004) find that in the 1964-1971 period, 20-25 percent of new female teachers were ranked in the top 10 percent of their high school cohort, while in 2000, less than 13 percent were ranked at the top decile. Hoxby and Leigh (2004) similarly find that the share of teachers in the highest aptitude category fell from 5 percent in 1963 to 1 percent in 2000 and the share in the lowest aptitude category rose from 16 percent to 36 percent in the same period.

[2] Merit pay faces opposition from the two major unions: The American Federation of Teachers (AFT) and the National Education Association (NEA). Though in favor of reforming teacher compensation systems, the AFT and the NEA officially object to programs that reward teachers based on student test scores and principal evaluations, while favoring instead systems that reward teachers based on additional roles and responsibilities they take within the school or certifications and qualifications they accrue. The AFT's official position cites the past underfunding of such programs, the confusing metrics by which teachers

empirical evidence on the effectiveness of such programs is ambivalent. In developing countries where the degree of teacher professionalism is extremely low and absenteeism is rampant, field experiments that link pay to teacher performance are associated with substantial improvements in student test scores (Duflo et al., 2012; Glewwe et al., 2010; Muralidharan and Sundararaman, 2011). On the other hand, the only two field experiments conducted in the United States have shown small, if not negative, treatment effects (Springer et al., 2010, Fryer, forthcoming).[3]

During the 2010-2011 school year we conducted an experiment in nine schools in Chicago Heights, IL. At the beginning of the school year teachers were randomly selected to participate in a pay-for-performance program. Among those who were selected the timing and framing of the reward payment varied significantly. One set of teachers – whom we label the "Gain" treatment – received "traditional" financial incentives in the form of bonuses at the end of the year linked to student achievement. Other teachers – the "Loss" treatment – were given a lump sum payment at the beginning of the school year and informed that they would have to return some or all of it if their students did not meet performance targets. Importantly, teachers in the "Gain" and "Loss" groups who had the same performance received the same final bonus.

Within the "Loss" and "Gain" groups we additionally test whether there are heterogeneous effects for individual rewards compared to team rewards. In all groups, we incentivized performance according to the "pay for percentile" method developed by Barlevy and Neal (2012), in which teachers are rewarded according to how highly their students' test score improvement ranks among peers with similar baseline achievement.

Students whose teachers were in the "Loss" treatment show large and statistically significant gains between 0.201σ (0.076) and 0.398σ (0.129) standard deviations in math test scores.[4] In line with previous studies in the United States, we do not find an impact of

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were evaluated, and the crude binary reward system in which there is no gradation of merit as the reasons for its objection. The NEA's official position maintains that any alterations in compensation should be bargained at the local level, and that a singular salary scale and a strong base salary should be the standard for compensation.

[3]Non-experimental analyses of teacher incentive programs in the United States have also shown no measurable success, though one should interpret these data with caution due to the lack of credible causal estimates (Glazerman et al., 2009; Vigdor, 2008)

[4] Our agreement with the Chicago Heights teachers union required us to offer every teacher the opportunity to participate in the incentive program, including Social Studies teachers, Writing teachers, Language Arts teachers, and interventionists. Since the district only administers Math, Reading, and Science tests (the last

teacher incentives that are framed as gains. The difference between the "Loss" and "Gain" treatments in math improvement is statistically significant.

Three additional pieces of data help provide a theoretical interpretation for the empirical results that we observe. One possibility is that the differences across treatment are spuriously driven by differential student attrition, but we find no evidence to support that hypothesis. Second, we consider whether our results could be driven by teacher cheating on the exams that determine the incentive pay. The fact that we obtain very similar results on non-incentivized statewide exams, however, argues against this possibility. Finally, we explore the extent to which relaxed credit constraints may have allowed the teachers paid in advance to invest more in classroom materials. In the data, however, teachers in the loss treatment report spending *less* money on classroom materials rather than more.

Consequently, we believe that the most plausible interpretation of our findings is that paying teachers in advance triggers loss aversion: giving back the money is more painful for the teachers than falling short of a bonus in the traditional *ex post* payment model. The psychological effect leads to a greater exertion of effort when teachers are given their bonuses in advance with the risk that those payments will be clawed back.

The remainder of the paper is organized as follows. Section 2 provides a brief literature review. Section 3 details the experiment and its implementation. Section 4 describes the data and research design used in the analysis. Section 5 presents estimates of the impact of teacher incentives on student achievement. Section 6 explores the mechanisms which might be at work in explaining our empirical findings. The final section concludes. There is an online data appendix that provides details on how we construct our covariates and our sample from the school district administrative files used in our analysis.

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only in 4th and 7th grade), we allowed Social Studies teachers, Writing teachers, Language Arts teachers, and reading interventionists to earn rewards based on their students' performance on the Reading examination. In other words, a student's reading performance often determined the rewards of multiple teachers, who were potentially assigned to different treatment groups. While this structure created some confusion among teachers and likely contaminated our Reading results, it allowed us to preserve a rigorous experimental design for our math treatments. In the interest of full disclosure, we present results for reading tests in the Appendix, but our discussion will focus primarily on the impacts of the various treatments on math performance. We discuss these issues in more detail in Sections 3 and 4.

II. A Brief Review of the Literature

The theory underlying teacher incentives programs is straightforward: if teachers lack motivation to put effort into important inputs to the education production function (e.g. lesson planning, parental engagement), financial incentives tied to student achievement may have a positive impact by motivating teachers to increase their effort.

There are a number of reasons, however, why teacher incentives may fail to operate in the desired manner. For instance, teachers may not know how to increase student achievement, the production function may have important complementarities outside their control, or the incentives may be either too confusing or too weak to induce extra effort. Moreover, if teacher incentives have unintended consequences such as explicit cheating, teaching to the test, or focusing on specific, tested objectives at the expense of more general learning, teacher incentives could have a negative impact on student performance (Holmstrom and Milgrom, 1991; Jacob and Levitt, 2003). Others argue that teacher incentives can decrease a teacher's intrinsic motivation or lead to harmful competition between teachers in what some believe to be a collaborative environment (Johnson, 1984; Firestone and Pennell, 1993).

Despite the controversy, there is a growing literature on the role of teacher incentives on student performance (Glazerman et al., 2009; Glewwe et al., 2010; Lavy, 2002; Lavy, 2009; Muralidharan and Sundararaman, 2011; Fryer, forthcoming, Springer et al., 2010; Vigdor, 2008.) including an emerging literature on the optimal design of such incentives (Neal, 2012). To date, there are five papers, three of them outside the US, which provide experimental estimates of the causal impact of teacher incentives on student achievement: Duflo et al. (2012), Glewwe et al. (2010), Muralidharan and Sundararaman (2011), Fryer (forthcoming) and Springer et al. (2010). Figure 1 displays the one-year treatment effects from these five experiments.

Evidence from Developing Countries

Duflo et al. (2012) randomly sampled 60 schools in rural India and provided them with financial incentives to reduce absenteeism. The incentive scheme was simple; teachers' pay was linear in their attendance, at the rate of Rs 50 per day, after the first 10 days of each month. They found that the teacher absence rate was significantly lower in

treatment schools (22 percent) compared to control schools (42 percent) and that student achievement in treatment schools was 0.17 σ (0.09) higher than in control schools.

Glewwe et al. (2010) report results from a randomized evaluation that provided 4th through 8th grade teachers in Kenya with group incentives based on test scores and find that while test scores increased in program schools in the short run, students did not retain the gains after the incentive program ended. They interpret these results as being consistent with teachers expending effort towards short-term increases in test scores but not towards long-term learning.

Muralidharan and Sundararaman (2011) investigate the effect of individual and group incentives in 300 schools in Andhra Pradesh, India and find that group and individual incentives increased student achievement by 0.165σ (.042) after one year. While the effects of the group incentive and the individual incentive treatments are very similar in year one, they diverge in the second year. Two-year effects are 0.283σ (.058) and 0.154 σ (0.057) for the individual and group treatments, respectively.

Evidence from Experiments in the United States

Springer et al. (2010) evaluate a three-year pilot initiative on teacher incentives conducted in the Metropolitan Nashville School System between the 2006-2007 school year and the 2008-2009 school year. Approximately 300 middle school mathematics teachers who volunteered to participate in the program were randomly assigned to the treatment or the control group, and those assigned to the treatment group could earn up to $15,000 as a bonus if their students made gains in state mathematics test scores equivalent to the 95th percentile in the district. They were awarded $5,000 and $10,000 if their students made gains equivalent to the 80th and the 90th percentiles, respectively. Springer et al. (2010) found there was no significant treatment effect on student achievement or on measures of teachers' response such as teaching practices.[5]

[5] There are several non-experimental evaluations of teacher incentive programs in the US, all of which report a non-significant impact of the program on student achievement. Glazerman et al. (2009) report a non-significant effect of -0.04 standard deviations on student test scores for the Teacher Advancement Program in Chicago and Vigdor (2008) reports a non-significant effect of the ABC School-wide Bonus Program in North Carolina. Outside the US, Lavy (2002, 2009) reports significant results for teacher incentive programs in Israel.

Fryer (forthcoming) conducted an experiment on teacher incentives in over 200 New York City public schools, which distributed a total of roughly $75 million to over 20,000 teachers. Each participating school could earn $3,000 for every union-represented staff member, which the school could distribute at its own discretion, if the school met the annual performance target set by the Department of Education based on school report card scores. Each participating school was given $1,500 per union staff member if it met at least 75% of the target, but not the full target. Each school had the power to decide whether all of the rewards would be given to a small subset of teachers with the highest value-added, whether the winners of the rewards would be decided by lottery, or virtually anything in-between. The only restriction was that schools were not allowed to distribute rewards based on job seniority. Yet, despite this apparent flexibility, the vast majority of schools chose to distribute the rewards evenly, and there was no effect on student achievement or teacher behavior. If anything, there was a negative impact, especially in larger schools where free-riding may have been problematic.

Our specific contribution is two-fold. First, this paper is the first experimental study to demonstrate that teacher merit pay can have a significant impact on student performance in the U.S. Second, we show that these findings differ from previous studies as a result of incorporating insights from behavioral economics into the incentive design. Moreover, we contribute to a small but growing literature that uses randomized field experiments to test incentive pay in organizations (Bandiera, Barankay, and Rasul 2007, Hossain and List, 2012). The paper that is in some ways most similar in spirit to this one is Hossain and List (2012), which uses a field experiment in a Chinese manufacturing plant to explore the effects of reward framing (but does not actually examine frontloading incentives).

III. Program Details

Background

The city of Chicago Heights is located thirty miles south of Chicago. The district contains nine K-8 schools with a total of approximately 3,200 students. Like larger urban school districts, Chicago Heights is made up of primarily low-income minority students

who struggle with low achievement rates—in the year prior to our program, 64% of students met the minimum standard on the Illinois State Achievement Test (ISAT) compared to 81% of students statewide. Roughly 98% of the elementary and middle school students in our sample are eligible for free or reduced-price lunch.

In cooperation with the Chicago Heights superintendent and Chicago Heights Teachers Union, the implementation of our field experiment followed standard protocols. As part of our agreement with the Teachers Union to conduct an experiment with teacher incentives, program participation was made available to every K-8 classroom teacher, as well as reading and math interventionists.[6] Approximately 160 teachers were eligible to participate in the experiment. After we introduced the program at the district-wide Teacher Institute Day at the start of the 2010-11 school year, teachers had until the end of September to opt-in to the program; 150 teachers (93.75%) elected to participate.

Table 1 provides a brief summary of the treatment. Participating teachers were randomly assigned to one of four treatment groups or a control group. Teachers in the incentive groups received rewards based on their students' end of the year performance on the ThinkLink Predictive Assessment, an otherwise low stakes standardized diagnostic assessment that is designed to be aligned with the high-stakes Illinois Standards Achievement Test (ISAT) taken by 3rd-8th graders in March (K-2 students take the Iowa Test of Basic Skills (ITBS) in March).[7] We calculate rewards using the "pay for percentile" methodology developed by Barlevy and Neal (2012). At baseline, we place each student in a bin with his nine nearest neighbors in terms of pre-program test performance.[8] We then rank each student within his bin according to improvement

[6] Interventionists pull students from class for 30-60 minutes of instruction in order to meet the requirements of Individualized Education Plans (IEPs) developed for students who perform significantly below grade level. All but one of the interventionists in Chicago Heights teaches reading.

[7] The results of ISAT are used to determine whether schools are meeting yearly targets under the No Child Left Behind law. The ThinkLink is administered to 3rd-8th grade students four times a year in September, November, January and May. K-2 students took the test in May only. Each subject test lasts 30-60 minutes and is either taken on the computer (3rd-8th grade students at all but one school) or on paper (all K-2 students and 3rd-8th grade students at one school). All students are tested in math and reading. In addition, 4th and 7th grade students take a science test as they do on ISAT. We proctored all end of the year testing in order to ensure consistency and discourage cheating.

[8] We construct a predicted baseline (fall 2010) test score based on up to three years of prior test score data. For students without prior year test scores, we use their actual fall 2010 score as their baseline score (fall testing was completed before the program began). Students are placed in separate bins for each subject. In order to avoid competition among teachers (or students) in the same school, students are never placed in a bin with students from the same school.

between his baseline and end of the year test score. Each teacher receives an "overall percentile", which is the average of all her students' percentile ranks within their bins.[9] Teachers receive $80 per percentile for a maximum possible reward of $8,000. The expected value of the reward ($4,000) is equivalent to approximately 8% of the average teacher salary in Chicago Heights.[10]

Random Assignment

Before any randomization occurs, we paired all teachers in each school with their closest match by grade, subject(s), and students taught. From there, the randomization procedure is straightforward for "contained" teachers – i.e. those who teach the same homeroom for the entire day. These teachers are randomly assigned to one of the four treatments, or the control group, subject to the restriction that teachers in the "team treatments" must be in the same treatment group as his/her teammate.

Teachers who teach multiple homerooms are subject to a slightly different procedure. Their students are grouped by class and subject and randomized into one of the five groups. As a result, a teacher can receive incentives based on the performance of some of her classes taught throughout the day and no incentives for others. In our main specifications, the students whose performance does *not* determine a teacher's reward are still included in the control group.

This aspect of our design allows us to examine the role of spillovers in our incentive scheme. If the teaching production has significant fixed costs, we might expect that students in the control group whose teacher is incentivized for *other* students' performance would benefit from the treatment. For instance, if a teacher teaches both a treated and a control math class, the added incentive to prepare an effective lesson for the treated class might result in a better lesson for control students.

One difficulty with the research design is that our agreement with the Chicago Heights Teachers Union required us to offer the incentive program to all elementary and

[9] We rewarded teachers of contained classrooms based on the performance of their homeroom on both reading and math (and science in 4th and 7th grades only). We rewarded teachers of rotating classrooms on a subset of the homeroom-subjects they taught.

[10] At the end of the year we rounded up students' percentiles to 100%, 90%, 80% . . . .20%, 10%, so that the average percentile was 55% (rather than 50%) and the average reward was $4,400 (rather than $4,000). Teachers were not informed of this in advance.

middle school teachers who signed up to participate. This presents few complications in math; students typically have only one math teacher, so there is a nearly one-to-one mapping between teachers and students. However, since the district does not administer exams in Social Studies, we offered incentives to these teachers based on their students' performance on the Reading exam.[11] Moreover, students with an Individualized Education Plan (IEP)—roughly 11% of the sample—also receive additional reading instruction from a reading specialist. Thus, more than one-third of the students in the sample have reading teachers in different treatments. Because of the confusion this likely induced among reading teachers and the difficulties that arise in the statistical analysis due to contamination and lack of power, we focus our discussion on the math results in what follows. Results for reading can be found in Appendix Tables 1 and 2.[12]

Our randomization procedure is straightforward. To improve balance among the control group and the four treatment arms, over a pure random draw, we re-randomized teachers after the initial draw. First, we calculated a balance statistic for the initial assignments, defined as the sum of the inverse p-values from tests of balance across all five groups.[13] Our algorithm then searches for teachers to "swap" until it finds a switch that does not violate any of the rules outlined above. If switching these teachers' treatments would improve the balance statistic, the switch is made; otherwise it is ignored. The algorithm continues until it has tested forty potential swaps.

There is an active discussion on which randomization procedures have the best properties. Treasure and MacRae (1998) prefer a method similar to the ones described above. Imbens and Wooldridge (2009) and Greevy et al. (2004) recommend matched pairs. Results from simulation evidence presented in Bruhn and McKenzie (2009) suggest that for large samples there is little gain from different methods of randomization over a pure single draw. For small samples, however, matched-pairs, re-randomization (the method employed here), and stratification all perform better than a pure random draw.

[11] At the request of the district, we also based incentives for Language Arts and Writing teachers on student performance on the Reading exam.

[12] We also incentivized 4th and 7th grade science, which are tested on the statewide exam. However, the sample sizes were too small to conduct a meaningful statistical analysis.

[13] We use chi-squared tests to test for balance across categorical variables (school and subject) and rank-sum tests for continuous variables (baseline ThinkLink math score, baseline ThinkLink reading score, contact minutes with teacher, percent black, percent Hispanic, percent female and grade).

Following the recommendation of Bruhn and McKenzie (2009), we have estimated our treatment effects including all variables to check balance. Whether we include these variables or a richer set of controls does not significantly alter the results.

Implementation

Teachers assigned to one of the two "Gain" treatments received their rewards at the end of the year, much like all previous programs have done (Springer et al., 2010; Fryer, forthcoming). In the "Loss" treatment, however, the timing changes significantly. Teachers in these treatment arms received $4,000 (i.e., the expected value of the reward) *at the beginning of the year*.[14] Teachers in the "Loss" treatment signed a contract stating that if their students' end of the year performance was below average, they would return the difference between $4,000 and their final reward.[15] If their students' performance was above average, we issued the teacher an additional payment of up to $4,000 for a total of up to $8,000. Thus, "Gain" and "Loss" teachers received identical net payments for a given level of performance. The only difference is the timing and framing of the rewards.

Within the "Gain" and "Loss" groups, teachers were also randomly assigned to receive either individual or team rewards. Teachers in the individual treatment groups received rewards based on the performance of their own students. The team treatment paired teachers in a school who were closely matched by grade and subject(s) taught. These teachers received rewards based on their average team performance. For example, if teacher A's overall percentile was 60% and teacher B's overall percentile was 40%, then their team average was 50% and each teacher received $4,000.[16]

The study formally commenced at the end of September 2010 after baseline testing was completed. Informational meetings for each of the incentive groups were held in October 2010, at which time the incentivized compensation was explained in detail to the teachers. Midway through the experiment, we provided incentivized teachers with an interim report summarizing their students' performance on a midyear assessment

[14] For tax reasons, some teachers requested that we issue the upfront payment in 2011. About half of teachers received the loss reward at the beginning of January 2011.
[15] We received back 98% of the payments owed at the end of the year.
[16] Ours is the first study to base rewards on teacher pairs. Previous studies have either tested individual or school-wide rewards. Muralidharan and Sundararaman (2011) compare individual and school-wide incentives in small schools averaging approximately 3 teachers each.

test (the results were for informational use only and did not affect teachers' final reward). We also surveyed all participating teachers about their time use, collaboration with fellow teachers and knowledge about the rewards program.

IV. Data and Research Design

Data

We merged administrative data from the Chicago Heights school district with survey data from the teachers in the experiment. The administrative data include: teacher assignments, gender, race, eligibility for free or reduced price lunch, eligibility for Special Education services, Limited English Proficiency (LEP) status, and student test scores. The test score data include: 2011 ISAT score (3rd-8th grade), 2011 ITBS score (K-2nd grade), 2010-2011 ThinkLink scores (baseline, fall, winter and spring), and up to 3 prior years of ISAT, ITBS and ThinkLink test scores (depending on the student's age and when he entered the district). The survey includes questions about program knowledge, collaboration with fellow teachers and time use. We received a 53% overall response rate (49% in the "Gain" group, 62% in "Loss" group and 36% in Control). We also worked with principals and teachers to confirm the accuracy of class rosters.

We use administrative data from 2007 through fall 2010 (pre-treatment) to construct baseline controls and spring 2011 test scores (post-treatment) for outcome measures. The main outcome variables are the direct outcomes we provide incentives for: spring ThinkLink scores in math for all K-8 students We also analyze the effect of treatment on Illinois's statewide exams taken in March (ISAT and ITBS).

Table 2 presents summary statistics for students in the "Gain" treatment, "Loss" treatment and control group.[17] Accounting for within-homeroom correlation, the groups are very well balanced. The table reinforces that our sample contains almost exclusively poor minority students; 98 percent are eligible for free or reduced-price lunch, and 96 percent are members of a minority group.

Econometric Approach

[17] Table 2 excludes students exposed to multiple treatments. These students are included in the regression results.

To estimate the causal impact of providing teacher incentives on outcomes, we estimate intent-to-treat (ITT) effects, i.e., differences between treatment and control group means. Let $GainTreat_h$ and $LossTreat_h$ be indicators for starting the year in a homeroom assigned to the Gain or Loss treatment, respectively. Furthermore, let X_i be a vector of baseline covariates measured at the individual level that includes all the variables summarized in Table 2. The individual level covariates, X_i, a set of school fixed-effects, γ_s, and set of grade fixed effects, μ_g, comprise our parsimonious set of controls. All these variables are measured pre-treatment, and the school and grade effects reflect a students' enrollment at the beginning of the school year.

The ITT effects, π_G and π_L, are estimated from the equation below:

$$achievement_i = \alpha + GainTreat_h \pi_G + LossTreat_h \pi_L + X_i \beta + \gamma_s + \mu_g + \varepsilon_{isgt} ,$$

where h indexes homerooms (the unit of randomization), i students, g grades, and s schools.

Recall, given our design, it is possible that a student has two or more teachers that face different incentive treatments within the same subject area. Because we focus on math teachers, this inconvenience is easily overcome; students have at most two math teachers and 99% of the students in our sample see a single math teacher. For students with one math teacher, the treatment variable is dichotomous and we estimate a traditional intent-to-treat regression. In the 1% of cases in which a student has two math teachers who are in different treatments, we assign him a value of 0.5 for *GainTreat* and 0.5 for *LossTreat*. Correspondingly, if one teacher is in control and one is in treatment, the relevant treatment variable takes on a value of 0.5. Dropping all students exposed to multiple teachers yields qualitatively identical results.[18]

V. Empirical Analysis

Table 3 presents ITT estimates of the effect of teacher incentives on math achievement measured by the May 2011 ThinkLink exam, the test used in determining

[18] The situation is significantly more complex for reading, where one-third of our sample is exposed to teachers in different treatment arms.

teacher incentive compensation in our study. We present estimated treatment effects in both standard deviation units and percentile ranks within groups of nine nearest neighbors – precisely the metric on which the rewards were calculated. The top three rows correspond to the "Loss" treatments, including the pooled loss results as well as the "Individual Loss" and "Team Loss" results separately. The bottom three rows present parallel results for the "Gain" treatments. Standard errors, clustered at the homeroom level, are presented in parentheses below each estimate.

Columns (1) and (4) include only a baseline test score as a control. Columns (2) and (5) add the remaining student-level variables in Table 2 as controls; their inclusion has almost no effect on our estimates. In columns (3) and (6), we include only treated and "pure control" students in the regression sample. In these specifications, we drop students whose teacher receives rewards based on other students' performance, but not these particular students. If our treatments create positive (resp. negative) spillovers onto these students, we would expect these estimates to be higher (resp. lower) than those in columns (2) and (5).

We find large impacts of the Loss treatment on ThinkLink math scores. Pooling the Team and Individual treatments together produces a treatment effect (with controls) of 0.220σ (0.065), or 6.840 (2.554) percentile points. When we drop control students at risk of receiving spillovers, our estimates are even higher: 0.338σ (0.101) and 9.559 (3.585) percentile points.

In contrast, similar to Fryer (forthcoming) and Springer et al. (2010), we find smaller, mostly insignificant coefficients in our pooled "Gain" treatments, in which teachers are rewarded at the end of the year for student achievement. The main pooled effect is 0.092σ (0.070), or 1.884 (2.843) percentile points. When we account for possible spillovers, the aggregate Gain treatment effect climbs to 0.185 (0.097) and is marginally significant. Importantly, we can reject the two-sided null hypothesis that Gain and Loss treatments have the same treatment effects with at least 95% confidence in all specifications.

Examining the individual and team treatment arms separately, the estimated effects of Team Loss are almost identical to the effects of Individual Loss. Similarly,

there is little of interest in the Individual Gain and Team Gain coefficients, except to note that they mirror the pooled results.

Table 4 presents an identical set of regressions that estimate the effect of student performance on statewide tests for which teachers did not receive any incentives. The similarities are striking: the pooled loss effect is 0.201σ (0.076) or 6.680 (3.194) percentile points. Accounting for potential spillovers, the coefficients increase to 0.398σ (0.129) and 10.838 (5.470) percentile points. The pooled gain effects are once again small and insignificant [-0.003σ (0.084) and 1.012 (3.351) percentile points], and quite similar in magnitude to the ThinkLink coefficients when we restrict the sample to pure control students [0.175σ (0.128), or 4.556 (5.198) percentile points].

Interestingly, the Team Loss estimates are consistently larger than the Individual Loss effects on the state tests. In percentile points, the team effect [10.803 (3.350)] dwarfs the individual effect [3.240 (3.917)], and we can reject the two-sided null hypothesis of equality with 99% confidence. The two effects are not statistically differentiable when we measure outcomes in scaled scores, however.

Similar tables for Reading can be found in the Appendix; Appendix Table 1 reports results for ThinkLink, and Appendix Table 2 reports ISAT and ITBS estimates. Just as in Math, the Team Loss treatment seems to have the strongest effects on reading scores, though the effects disappear when combined with the negative Individual Loss effects in the pooled specifications. The spillover-adjusted estimates are also higher than the unadjusted estimates in all regressions. While large standard errors prevent definitive conclusions, we view this as further suggestive evidence that our treatments had beneficial effects for students in certain control classrooms.

To investigate the heterogeneity of the program's effects, Table 5A presents results split out across genders, races, and baseline test performance. Because of the large number of coefficients presented, we show results only for the percentile improvement or decline on ThinkLink.[19] Column 1 presents results for the full sample, repeating the estimates shown in Table 3. The next two columns break down the results by gender. The following two columns divide the sample by race, and the last two columns according to whether a student's baseline test score was above or below the

[19] Full results are available from the authors on request.

median baseline score in his grade. There is little systematic heterogeneity that we are able to detect. In no case is the comparison of treatment effects statistically different across the sub-groups.

An interesting pattern emerges when we separate students by grade level however. As Table 5B demonstrates, each treatment has large and statistically significant effects among students in second grade and below – ranging from 14.487 (6.403) percentile points in Team Gain to 17.437 (6.126) in Individual Loss. Among 3rd-8th graders, the effects are more muted, and only the loss treatment effects are differentiable from zero. Pooling the loss treatments yields a treatment effect of 5.791 (2.555) percentile points among older students, as opposed to -0.740 (3.266) in pooled gain.

VI. Interpretation

The results presented thus far suggest that the timing of bonus payments can play an important role in amplifying the power of teacher incentives. When we offer teachers performance rewards paid at the end of the year, we replicate the null findings in Springer et al. (2010) and Fryer (forthcoming). However, when we provide an advance on the payment and re-frame the incentive as avoidance of a loss, we observe treatment effects in excess of 0.201σ and some as high as 0.398σ—similar to the increase in student achievement associated with a more than one standard deviation increase in teacher quality. In all of our specifications, we can reject the null hypothesis of equal effects in the Loss and Gain treatments with 95% confidence.

There are a number of potential explanations for the observed divergence of results. One possibility is that the findings are spurious – driven by differential student attrition or teacher cheating. A second possibility is that the results are real and that the source of the improvement operates through the relaxation of liquidity constraints, allowing teachers to invest more in classroom materials. A third explanation is that teachers doubt the veracity of the promises to pay the incentive; the provision of upfront money gives the program credibility and thus induces greater effort. A final possibility is that providing the money in advance and requiring the teachers to repay if their students underperform exploits loss aversion, inducing greater effort from the teachers. We address these alternative hypotheses one by one below.

Attrition

It is possible that teachers in the loss treatment resorted to means other than teaching to increase their rewards. One might worry that they found ways to discourage their weaker students from taking exams, for instance. The simplest version of this strategy would not be effective in our incentive structure since rewards are based on student performance *relative* to other students with similar baseline. Teachers could, however, potentially weed out students who have made relatively less progress during the year.

To investigate the prevalence of attrition in our sample, we run a probit regression on all of our covariates where the dependent variable is an indicator for missing any of the exams we consider. Table 6 reports the marginal effects of our Gain and Loss treatments, evaluated at the mean.

There is little evidence that differential attrition is influencing our results. Students in the Gain treatments are 2.1 (1.3) percentage points less likely than the control group to be missing ThinkLink Math scores, but the difference is not statistically significant. Similarly, they are more likely to be missing ISAT/ITBS scores by roughly the same magnitude – 2.8 (2.0) percentage points in Math. The Loss treatment demonstrates smaller and insignificant results. This is important because if a student misses a state exam, a score of zero is imputed when calculating the school-wide average. Given that we see almost identical treatment effects across ThinkLink and the state exams for which our teachers were not directly incentivized, it seems unlikely that attrition is driving our results.

Cheating

Tying bonuses to test scores might induce certain teachers to cheat. Indeed, Jacob and Levitt (2003) find an uptick in estimated cheating after Chicago Public Schools instituted a performance-bonus system. The best argument against cheating in this context is that the results on the state tests – for which teachers received no rewards under our scheme and for which the entire school was under pressure to perform – mirror the ThinkLink results. It seems unlikely that the pattern would repeat itself on the ITBS and

ISAT tests if differential cheating practices across treatment and control groups were driving our results.[20]

Relaxing the liquidity constraint

It is also possible that front-loading enables teachers to make productivity-enhancing investments in their classroom (say, new workbooks or dry-erase markers). With perfect credit markets, we would not expect any difference in effects between our gain and loss treatments. If teachers are liquidity-constrained, however, the loss treatment effectively gives them access to a form of financing unavailable to teachers in the gain treatment.[21]

Survey evidence, however, does not support this explanation. Table 7 reports treatment effect estimates on several survey outcomes. When asked how much of their personal money they spent on classroom materials that year, teachers in the loss group reported spending less than teachers in the gain treatment or the control group (though the difference is not statistically significant).

Equally damaging to a liquidity constraint hypothesis is that 69% of teachers in the loss treatment report that they had not spent any money from their check when they were surveyed in March 2011 (three quarters of the way through the experiment).

Trust

It is possible that the teachers did not fully trust the experimenters to fulfill the agreement set out at the beginning of the school year. Paying some portion of the reward up front might build trust among the teachers. While we cannot directly rule out this possibility, two pieces of evidence work against it. First, in advance of this experiment, the researchers had a long history of positive and mutually beneficial engagement with this school district. The teacher's union and district administration were heavily involved in the design, implementation, and roll-out of the experiment. Second, teachers in both

[20] Additionally, the way in which ThinkLink is administered makes teacher cheating difficult. ThinkLink is administered in a computer lab, not the usual classroom, and the research team proctored all exams.

[21] Even absent liquidity constraints, it is possible that frontloading could increase classroom investment if teachers suffer from mental accounting. Teachers might treat upfront incentive money as a separate account than their other savings and might have a greater propensity to spend the incentive money on classroom supplies.

the "Loss" and "Gain" treatments would likely have been aware of the upfront payments to the "Loss" teachers. Thus, to the extent that early payments increased credibility, it likely did so to both treatment groups.[22]

Loss Aversion

Absent compelling support for the various hypotheses discussed above, we view the most plausible explanation for our finding is the loss-aversion inducing frame of the upfront payment. Beginning with the seminal work of Kahneman and Tversky (1979), a large body of research has been amassed documenting the power of loss aversion to motivate individual behavior in the lab. Experiments have consistently demonstrated that subjects are more responsive in settings framed as losses relative to those framed as gains. Providing the incentive pay upfront has the potential to shift the recipients frame in a powerful way: Giving back the cash advance *feels* very different to people than not getting paid a bonus, even though ultimately the same financial end result is achieved.

One important consideration is whether we should expect these types of gains in the long run or if teachers will become accustomed to these influences and the effect will dull through time. For our purposes, the results in List (2003, 2004) are the most relevant, since they show that market experience has an important influence on the shape and magnitude of the disparity. In particular, List reports that market experience measured by activity in naturally-occurring trading markets for pins, sports card, and sports memorabilia leads to behavior that is not significantly different from neoclassical expectations. But, importantly, a considerable amount of market experience is necessary before the loss aversion effects begin to dampen: dozens of trades over several months are necessary to dampen the influence of loss aversion.

[22] A more likely scenario is that the teachers who received up-front payments would doubt the ability of the researchers to claw back the money if the students underperformed. (Indeed, prior to initiating the experiment there was a great deal of discussion among the principal investigators and the University legal team regarding the ability to enforce the "Loss" contracts.) If "Loss" teachers felt that we could not take back their advances, they would face much weaker incentives to exert effort, and we would expect "Gain" to outperform "Loss."

VII. Conclusion

Recently a number of school districts around the U.S. have implemented standard teacher incentive programs, backloading merit pay to increase teacher effort and move student achievement. Past studies have found that such reform attempts have limited effects on student achievement. In this study, we present evidence that framing a teacher incentive program in terms of losses rather than gains leads to improved student outcomes. The impacts we observe are large – roughly the same order of magnitude as increasing average teacher quality by more than one standard deviation.

While there is empirical evidence from the lab that rewards framed as losses are more effective than rewards framed as gains in such settings, there have been few prior field experimental demonstrations of this phenomenon in naturally occurring principal-agent settings (but see Hossain and List, 2012 and List and Savikhin, 2013). It should be noted, however, that even if frontloading incentives leads to better student performance, it does not immediately follow that such an approach dominates the standard, ex post approach to compensation. One reason is that the teachers themselves might experience significant disutility from the incentive contract. Even though neither our teachers nor the factory workers in Hossain and List (2012) reveal adverse labor supply effects during or after the experiment, there is evidence in our own on-going work that suggests attrition can be important. Using a field experiment with workers on farms in the Midwest who tassel corn, pick fruits and vegetables, and bail hay, we find evidence that seasonal workers who were given a loss contract were less likely to return to work the subsequent summer than those given a standard gain contract. Clearly, much more work needs to be completed to make a definitive statement on the welfare properties of such incentive schemes.

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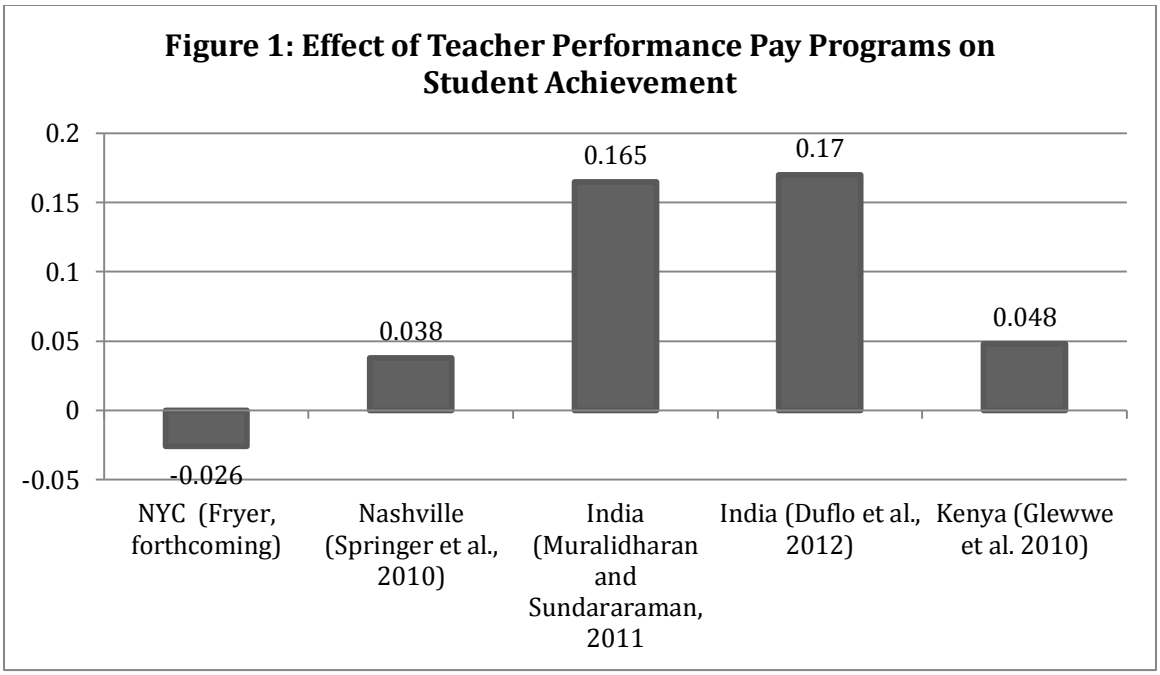
Figure 1: Effect of Teacher Performance Pay Programs on Student Achievement
0.2
0.15
0.1
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-0.05
-0.026
0.038
0.165
0.17
0.048
NYC (Fryer, forthcoming)
Nashville (Springer et al., 2010)
India (Muralidharan and Sundararaman, 2011
India (Duflo et al., 2012)
Kenya (Glewwe et al. 2010)

Table

Table 1: Summary of Teacher Incentive Program

Panel A: Overview		
Schools	Nine K-8 schools in Chicago Heights, IL	
Operations	$632,960 distributed in incentive payments, 90% consent rate.	
Panel B: Outcomes of Interest		
	Subjects and Grades	Date of Assessment
ThinkLink Learning Diagnostic Assessment	Math (K-8th), Reading (K-8th), and Science (4th and 7th)	May 2011
Illinios Standards Achievement Test	Math (3rd-8th), Reading (3rd-8th), and Science (4th and 7th)	March 2011
Iowa Test of Basic Skills	Math (K-2nd) and Reading (K-2nd)	March 2011
Panel C: Treatment Details		
	Timing of Rewards	Basis For Rewards
Individual Loss	Teachers Receive $4,000 check in September 2010; must pay back difference in June 2011	Teacher's own students
Individual Gain	Teachers paid in full in June 2011	Teacher's own students
Team Loss	Teachers Receive $4,000 check in September 2010; must pay back difference in June 2011	Teacher's and teammate's students
Team Gain	Teachers paid in full in June 2011	Teacher's and teammate's students
All Treatments	Treated teachers earned between $0 and $8,000 in bonus payments based on students' performance relative to nine statistically similar students in one of the other eight schools. Rewards are linear in a student's rank, so the expected value of the reward is $4,000.	

Table 2: Summary Statistics by Treatment Arm

	Control	Gain	Loss	*p-val*
Female	0.501	0.489	0.484	0.791
White	0.043	0.048	0.044	0.957
Black	0.357	0.359	0.365	0.978
Hispanic	0.598	0.573	0.576	0.953
Free or Reduced Price Lunch	0.988	0.981	0.972	0.172
Limited English Proficiency	0.168	0.151	0.121	0.866
Special Education	0.121	0.096	0.121	0.440
Baseline Thinklink Math Score	-0.069	0.059	0.037	0.451
Baseline Thinklink Reading Score	-0.040	0.040	0.021	0.722
Observations	656	981	982	
Homerooms	38	48	59	

Notes: This table presents summary statistics for the control group and our two treatment arms. Thinklink scores are standardized to have mean zero and standard deviation one. Column (4) displays a p-value from a test of equal means in the three groups, with standard errors clustered at the homeroom level.

Table 3: The Effect of Treatment on Thinklink Math Scores (ITT)

	Percentile Rank			*Scaled Score*		
	(1)	(2)	(3)	(4)	(5)	(6)
Pooled Loss	6.866**	6.840***	9.559***	0.222***	0.220***	0.338***
	(2.677)	(2.554)	(3.585)	(0.070)	(0.065)	(0.101)
Individual Loss	5.827*	6.284**	9.128**	0.212**	0.220***	0.346***
	(3.090)	(3.002)	(4.114)	(0.081)	(0.078)	(0.117)
Team Loss	8.201***	7.471**	10.011***	0.238***	0.221***	0.335***
	(3.111)	(2.889)	(3.690)	(0.084)	(0.076)	(0.102)
Pooled Gain	1.263	1.884	3.901	0.078	0.092	0.185*
	(2.888)	(2.843)	(3.508)	(0.072)	(0.070)	(0.097)
Individual Gain	0.942	1.939	3.706	0.066	0.092	0.178*
	(3.299)	(3.293)	(3.889)	(0.083)	(0.081)	(0.106)
Team Gain	1.777	1.846	4.070	0.095	0.093	0.196*
	(3.440)	(3.343)	(3.891)	(0.085)	(0.082)	(0.104)
Controls?	N	Y	Y	N	Y	Y
Spillovers Removed?	N	N	Y	N	N	Y
Pr(Gain = Loss)	0.019	0.031	0.015	0.019	0.027	0.008
Observations	2311	2311	1986	2311	2311	1986
Clusters	141	141	125	141	141	125

Notes: This table presents estimates of the effectiveness of various teacher incentive structures on Thinklink math scores. Results are shown for two outcome measures: a student's percentile rank on an end-of-year test relative to her nine nearest neighbors, and her score on that test normalized to have mean zero and standard deviation one within each grade. All regressions include dummy variables for each students' school and grade as well as baseline test scores. Columns (2), (3), (5), and (6) add controls for students' race, gender, age, free-lunch status, English proficiency, and special education status. To minimize spillover effects, columns (3) and (6) and drop students whose teachers were rewarded based on the performance of other students in the sample. We also present a p-value on the null hypothesis of equal treatment effects in pooled loss and pooled gain. Standard errors are clustered at the homeroom level throughout. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Table 4: The Effect of Treatment on ISAT and ITBS Math Scores (ITT)

	Percentile Rank			*Scaled Score*		
	(1)	(2)	(3)	(4)	(5)	(6)
Pooled Loss	6.867**	6.680**	10.838**	0.213***	0.201***	0.398***
	(3.269)	(3.194)	(5.470)	(0.078)	(0.076)	(0.129)
Individual Loss	2.915	3.240	7.333	0.157*	0.149	0.357**
	(3.918)	(3.917)	(6.311)	(0.090)	(0.092)	(0.146)
Team Loss	11.686***	10.803***	14.025**	0.292***	0.271***	0.444***
	(3.472)	(3.350)	(5.564)	(0.092)	(0.085)	(0.133)
Pooled Gain	0.228	1.012	4.556	-0.013	-0.003	0.175
	(3.402)	(3.351)	(5.198)	(0.085)	(0.084)	(0.128)
Individual Gain	0.205	0.991	3.883	-0.052	-0.036	0.137
	(3.818)	(3.770)	(5.346)	(0.098)	(0.096)	(0.134)
Team Gain	0.666	1.207	4.569	0.042	0.039	0.211
	(4.283)	(4.251)	(6.007)	(0.115)	(0.113)	(0.154)
Controls?	N	Y	Y	N	Y	Y
Spillovers Removed?	N	N	Y	N	N	Y
Pr(Gain = Loss)	0.017	0.042	0.025	0.003	0.006	0.004
Observations	2144	2144	1828	2144	2144	1828
Clusters	140	140	124	140	140	124

Notes: This table presents estimates of the effectiveness of various teacher incentive structures on ISAT/ITBS math scores. Results are shown for two outcome measures: a student's percentile rank on an end-of-year test relative to her nine nearest neighbors, and her score on that test normalized to have mean zero and standard deviation one within each grade. All regressions include dummy variables for each students' school and grade as well as baseline test scores. Columns (2), (3), (5), and (6) add controls for students' race, gender, age, free-lunch status, English proficiency, and special education status. To minimize spillover effects, columns (3) and (6) and drop students whose teachers were rewarded based on the performance of other students in the sample. We also present a p-value on the null hypothesis of equal treatment effects in pooled loss and pooled gain. Standard errors are clustered at the homeroom level throughout. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Table 5A: The Impact of Treatment on Math Percentile Scores Within Demographic Subgroups

	Whole Sample	*Gender* Male	Female	p-val	*Race* Black	Hispanic	p-val	*Baseline Score* Above Median	Below Median	p-val
Pooled Loss	6.840***	5.942**	8.681***		10.181**	9.531***		7.421**	5.718*	
	(2.554)	(2.893)	(3.230)	0.404	(4.029)	(3.183)	0.892	(3.484)	(3.346)	0.704
Individual Loss	6.284**	6.624*	7.516**		9.388**	10.354***		4.914	7.761**	
	(3.002)	(3.388)	(3.696)	0.805	(4.167)	(3.783)	0.841	(4.066)	(3.652)	0.565
Team Loss	7.471**	5.424	10.001***		11.027**	8.894**		11.338***	3.403	
	(2.889)	(3.481)	(3.573)	0.250	(5.337)	(3.683)	0.727	(3.636)	(3.873)	0.097
Pooled Gain	1.884	2.993	2.028		0.449	5.498		2.567	0.256	
	(2.843)	(2.836)	(3.640)	0.759	(3.397)	(3.333)	0.241	(3.920)	(3.343)	0.618
Individual Gain	1.939	2.306	2.662		0.274	4.440		1.363	1.914	
	(3.293)	(3.274)	(4.231)	0.917	(4.179)	(3.996)	0.410	(4.499)	(3.927)	0.916
Team Gain	1.846	3.892	1.372		0.504	6.588		4.247	-1.714	
	(3.343)	(3.578)	(4.008)	0.494	(4.278)	(3.971)	0.180	(4.215)	(4.213)	0.256
Observations	2311	1155	1122		584	941		1225	1078	
Clusters	141	141	135		95	102		134	139	

Notes: This table reports treatment effects for various subgroups in the data. The outcome variable is the student's percentile rank within a group of her nine nearest neighbors. Columns (5), (8), and (11) report p-values resulting from a test of equal coefficients between the gender, race, and baseline test score groups, respectively. Standard errors (clustered at the homeroom level) are reported in parentheses. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Table 5B: The Impact of Treatment on Math Percentile Scores Within Grade Level Subgroups

	Whole Sample	*Grade Level* K-2	3-8	p-val
Pooled Loss	6.840***	17.745***	5.791**	
	(2.554)	(6.126)	(2.555)	0.065
Individual Loss	6.284**	17.437**	6.603**	
	(3.002)	(6.922)	(2.871)	0.137
Team Loss	7.471**	17.328**	5.690*	
	(2.889)	(6.931)	(3.157)	0.118
Pooled Gain	1.884	15.515***	-0.740	
	(2.843)	(5.359)	(3.266)	0.008
Individual Gain	1.939	15.859***	-3.286	
	(3.293)	(5.247)	(4.244)	0.004
Team Gain	1.846	14.487**	2.544	
	(3.343)	(6.403)	(3.538)	0.093
Observations	2311	638	1673	
Clusters	141	45	97	

Notes: This table reports treatment effects for grade-level subgroups in the data. The outcome variable is the student's percentile rank within a group of her nine nearest neighbors. The rightmost column reports the p-values resulting from a test of equal coefficients between the two groups. Standard errors (clustered at the homeroom level) are reported in parentheses. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Table 6: Attrition

	Gain	Loss
Missing Thinklink Math Score	-0.021	0.006
	(0.013)	(0.017)
	N = 2511	
Missing Thinklink Reading Score	-0.024	-0.001
	(0.014)	(0.013)
	N = 2419	
Missing ISAT Math Score	0.028	0.001
	(0.020)	(0.022)
	N = 2511	
Missing ISAT Reading Score	0.027	0.010
	(0.023)	(0.025)
	N = 2419	

Notes: This table presents the increase in the probability of several measures of attrition associated with our gain and loss treatments. The results shown are the marginal effects calculated from a probit regression of the relevant dependent on treatment indicators and our full list of control variables. Standard errors (in parentheses) are clustered at the homeroom level. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Table 7: Teacher Survey Results

Outcome	Gain	Loss
Hours Lesson Planning	0.026	-0.461
	(1.256)	(1.258)
Hours Grading	-0.328	-1.647
	(1.438)	(1.440)
Hours Calling or Meeting w/ Parents	0.138	-0.071
	(0.458)	(0.459)
Hours Tutoring Outside of Class	1.092	0.953
	(1.742)	(1.744)
Hours Leading Extracurricular Activities	0.585	0.555
	(1.330)	(1.332)
Hours Completing Administrative Work	-1.587*	-1.258
	(0.936)	(0.938)
Hours Completing Professional Development Coursework	0.464	0.084
	(1.392)	(1.394)
Personal Money Spent on Class Materials ($)	19.744	-77.895
	(101.722)	(101.874)

Notes: This table presents results gathered from surveys of teachers in our experimental group at the end of the school year. All coefficients are derived by regression the outcome variable in the first column on a dummy for participation in the Gain or Loss treatment. The sample size for each regression is 82 teachers. Teachers are considered to have participated in either type of treatment if they receive that type of incentive based on any of their students' performance. Standard errors are reported in parentheses. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Appendix Table 1: The Effect of Treatment on Thinklink Reading Scores (ITT)

	Percentile Rank			*Scaled Score*		
	(1)	(2)	(3)	(4)	(5)	(6)
Pooled Loss	-0.006	0.266	4.649	0.006	0.018	0.179
	(2.778)	(2.761)	(4.093)	(0.067)	(0.064)	(0.115)
Individual Loss	-8.060**	-7.875**	-4.047	-0.189**	-0.174*	-0.028
	(3.739)	(3.796)	(4.268)	(0.095)	(0.090)	(0.118)
Team Loss	5.640**	5.945**	13.637***	0.139**	0.150**	0.388***
	(2.795)	(2.814)	(4.450)	(0.064)	(0.065)	(0.125)
Pooled Gain	-0.095	-0.082	6.061	-0.030	-0.036	0.159
	(2.649)	(2.581)	(3.693)	(0.063)	(0.060)	(0.104)
Individual Gain	2.708	2.656	10.647***	0.054	0.044	0.285**
	(3.381)	(3.360)	(4.031)	(0.082)	(0.080)	(0.116)
Team Gain	-5.451*	-5.395*	1.585	-0.182**	-0.182**	0.027
	(3.095)	(3.051)	(4.106)	(0.080)	(0.078)	(0.112)
Controls?	N	Y	Y	N	Y	Y
Spillovers Removed?	N	N	Y	N	N	Y
Pr(Gain = Loss)	0.974	0.899	0.604	0.632	0.464	0.779
Observations	2242	2242	1848	2242	2242	1848
Clusters	140	140	119	140	140	119

Notes: This table presents estimates of the effectiveness of various teacher incentive structures on Thinklink reading scores. Results are shown for two outcome measures: a student's percentile rank on an end-of-year test relative to her nine nearest neighbors, and her score on that test normalized to have mean zero and standard deviation one within each grade. All regressions include dummy variables for each students' school and grade as well as baseline test scores. Columns (2), (3), (5), and (6) add controls for students' race, gender, age, free-lunch status, English proficiency, and special education status. To minimize spillover effects, columns (3) and (6) and drop students whose teachers were rewarded based on the performance of other students in the sample. We also present a p-value on the null hypothesis of equal treatment effects in pooled loss and pooled gain. Standard errors are clustered at the homeroom level throughout. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

Appendix Table 2: The Effect of Treatment on ISAT/ITBS Reading Scores (ITT)

	Percentile Rank			*Scaled Score*		
	(1)	(2)	(3)	(4)	(5)	(6)
Pooled Loss	-4.102	-3.848	-0.222	0.008	0.026	0.079
	(3.285)	(3.194)	(4.937)	(0.085)	(0.075)	(0.115)
Individual Loss	-8.844**	-8.088*	-4.878	-0.096	-0.074	-0.039
	(4.358)	(4.310)	(5.474)	(0.111)	(0.099)	(0.128)
Team Loss	-0.792	-0.924	4.813	0.078	0.093	0.206
	(3.935)	(3.837)	(5.223)	(0.107)	(0.096)	(0.125)
Pooled Gain	-2.218	-2.526	1.272	-0.004	-0.022	0.033
	(2.869)	(2.816)	(4.210)	(0.076)	(0.070)	(0.106)
Individual Gain	-0.872	-1.039	4.522	0.043	0.027	0.122
	(3.352)	(3.243)	(4.139)	(0.086)	(0.076)	(0.100)
Team Gain	-5.167	-5.586	-1.988	-0.095	-0.115	-0.059
	(3.470)	(3.548)	(4.655)	(0.100)	(0.095)	(0.123)
Controls?	N	Y	Y	N	Y	Y
Spillovers Removed?	N	N	Y	N	N	Y
Pr(Gain = Loss)	0.556	0.675	0.637	0.886	0.541	0.579
Observations	2347	2347	1919	2347	2347	1919
Clusters	144	144	123	144	144	123

Notes: This table presents estimates of the effectiveness of various teacher incentive structures on ISAT and ITBS reading scores. Results are shown for two outcome measures: a student's percentile rank on an end-of-year test relative to her nine nearest neighbors, and her score on that test normalized to have mean zero and standard deviation one within each grade. All regressions include dummy variables for each students' school and grade as well as baseline test scores. Columns (2), (3), (5), and (6) add controls for students' race, gender, age, free-lunch status, English proficiency, and special education status. To minimize spillover effects, columns (3) and (6) and drop students whose teachers were rewarded based on the performance of other students in the sample. We also present a p-value on the null hypothesis of equal treatment effects in pooled loss and pooled gain. Standard errors are clustered at the homeroom level throughout. *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels, respectively.

ONLINE APPENDIX – NOT FOR PUBLICATION

A. Variable Construction

Attrition
For our experiment, we agreed to base teacher rewards on students' performance relative to other students of similar ability. Our experimental sample therefore includes all students for whom we observe enough baseline information to compute a "predicted" score at the time of treatment assignment. In other words, for a student to enter our experimental sample, we must observe either (a) a baseline ThinkLink score from Fall 2010 or (b) a full set of test scores from the pre-treatment year. Students in this group for whom we do not observe a final test score are considered attriters.

Students who enter the district in the middle of the year are not included in our sample.

Free and Reduced Price Lunch
We construct an indicator variable set to one if a student meets guidelines for free or reduced price lunch, as determined by District administrative files.

Limited English Proficiency
We set the LEP indicator to one if the District considers a student to be a "Limited English Speaker" or a "Non English Speaker" and zero if the student is a "Fluent English Speaker" or "Native English Speaker."

Race/Ethnicity
We code the race variables such that the five categories – white, black, Hispanic, Asian, and other – are complete and mutually exclusive. Hispanic ethnicity is an absorbing state. Hence "white" implies non-Hispanic white, "black" non-Hispanic black, and so on.

Special Education
We construct an indicator variable for students who receive special instruction according to an Individualized Education Plan, as recorded by the District.

Test Scores
For grades K-8, we observe scaled Math and Reading scores on four ThinkLink examinations taken during the 2010-11 school year (baseline, fall, winter, and spring). Teachers' rewards were determined by their student's performance on the final test, and it is our main outcome variable.

Students also take the Iowa Test of Basic Skills (grades K-2) and the Illinois Standard Achievement Test (grades 3-8). We observe math and reading scores for all grades. Where available, we use scores from the Spring 2010 exams as control variables.

To simplify interpretation, we normalize all scores to have mean zero and standard deviation one by grade, subject, and test. When scores are used as control variables, we leave them un-standardized and use the scale score interacted with the student's grade level.

Treatment Variables

To compute ITT effects, all treatment designations are based on the teacher(s) to which a student is initially assigned. When a student has multiple teachers incentivized on a given test, we assign treatment values according to the dosage assumption described in the text.